

Cart service known errors

Name	Add-to-Cart Failures — Inventory Service Timeout — cart-api
Status	ACTIVE
Version	1.1 (see revision note below)
Owner	cart-platform
Related errors	Cart Session Expiry — cart-api (different alert trigger, distinct cause — see Related errors note)
Summary	Add-to-cart requests fail with a 502/504 as a step change (not a gradual climb) when inventory-svc stops responding within cart-api's client timeout. cart-api's own database and resource metrics stay normal throughout, pointing to the downstream dependency rather than a local resource constraint.

Error details

Description cart-api calls inventory-svc synchronously on every add-to-cart request to confirm stock before confirming the item to the client. When inventory-svc's response exceeds cart-api's configured `client_timeout_ms`, cart-api aborts the call and returns a 502/504 to the caller. Because the failure is a hard timeout rather than a resource ceiling being approached, the error rate jumps as a step change at the moment inventory-svc degrades, with no preceding latency ramp.

Impacted services

cart-api (add-to-cart flow only). Not impacted: checkout-api, payment-gateway-svc, fraud-scoring-svc — checkout-api reads from a cached inventory snapshot taken at cart-finalization time and does not call inventory-svc directly, so this error does not propagate into the checkout path.

Primary configuration item

Client timeout: `cart-api/config/inventory-client.yaml` (`timeout_ms`, `retry_count`). inventory-svc's own SLA target: `p99 < 300ms` (see inventory-svc dashboard).

Steps to reproduce

Not deterministically reproducible on demand. Historically correlates with inventory-svc's own deploys or GC-pause events; has not been reproduced by cart-api load alone.

Workaround

Temporarily raise cart-api's inventory-client `timeout_ms` via the dynamic config flag (`inventory_client_timeout_override`) to reduce false-positive failures while inventory-svc is investigated — this is a runtime flag flip, not a deploy. Confirm with inventory-svc's on-call before raising it, since a longer timeout also means slower failure detection if inventory-svc is fully down rather than degraded.

Solution

No permanent structural fix has shipped yet. Tracked: add a circuit breaker so cart-api fails fast to a “confirm availability at checkout” fallback message instead of blocking add-to-cart on inventory-svc (org/cart-api#118), and move the stock check off the request path entirely via an eventually-consistent local cache (org/cart-api#121).

Revision note (added, not in original playbook): version bumped to 1.1 after the workaround changed from “retry inventory-svc 3x” (which compounded inventory-svc load during its own incidents) to a timeout-override flag. Last reviewed 2026-07-14.